PassFrame tuition

Letter Sequences & Alphabet Code

Year 5 Accelerated — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. E (+1 each time)

2. N (+1 each time)

3. M (+2 each time)

4. O (+3 each time)

5. V (−1 each time)

6. L (−2 each time)

7. I (F is the 6th letter; 6+3=9=I)

8. G (K is the 11th letter; 11−4=7=G)

9. I (+2 each time)

10. Q (+1 each time)

11. EPH (D→E, O→P, G→H)

12. Q (+3 each time)

13. Q (−2 each time)

14. D (B=2, F=6, halfway point = 4 = D)

15. P (+2 each time)

Standard

1. J (+3 each time, wrapping from Z back round to A)

2. R (−3 each time, wrapping from A back round to Z)

3. MTWXJ (H→M, O→T, R→W, S→X, E→J)

4. A (R is the 18th letter; 18+9=27, which wraps round to 1 = A)

5. S (C is the 3rd letter; 3−10=−7, which wraps round to S)

6. B (the 24th letter is X; X+4 wraps round to B)

7. H (+3 each time, wrapping from Z back round to A)

8. GOLD (encoding checks out: G→K, O→S, L→P, D→H, giving KSPH, so decoding KSPH gives back GOLD)

9. M (the gaps go +2, +3, +2, +3, so the next gap is +2: K + 2 = M)

10. K (E=5, Q=17, halfway = 11 = K)

11. NLFHV (M ↔ N, O ↔ L, U ↔ F, S ↔ H, E ↔ V)

12. MOUSE (the reverse-code swap undoes itself)

13. Q (+5 each time, wrapping from Z back round to A)

14. D (30 wraps round: 30−26=4 = D)

15. Q (the code is its own reverse, so the letter that swaps to J is the same letter J swaps to: Q)

Standard / Hard

1. X (two patterns woven together: front A, B, C… and back Z, Y, X…)

2. P (the gap grows by 1 each time: +2, +3, +4, +5)

3. F (+2 each time, wrapping from Z back round to B)

4. RGCT

5. MICE

6. N (gaps are +2, +2, then +3, +3, +3)

7. HPE (reverse to GOD, then G→H, O→P, D→E)

8. DCE (C+1=D, A+2=C, B+3=E)

9. U (1st, 3rd, 5th letters go A, C, E, +2 each; 2nd, 4th, 6th letters go Y, W, ?, −2 each)

10. W (D is 3 places after A; 3 places before Z is W)

11. L (5+7=12=L)

12. ENH (D→E, O→N, G→H)

13. CF (Z→C, C→F)

14. M (the next gap is 5, following the pattern 1, 1, 2, 3, 5; H + 5 = M)

15. W (C+1=D, reverse of D is W)

Homework — Answers

Easy / Standard

1. F (+1 each time)

2. O (+1 each time)

3. L (+2 each time)

4. R (+4 each time)

5. U (−1 each time)

6. K (−2 each time)

7. J (H=8; 8+2=10=J)

8. K (P=16; 16−5=11=K)

9. N (+2 each time)

10. R (+1 each time)

11. DBU (C→D, A→B, T→U)

12. P (+3 each time)

13. O (−2 each time)

14. F (D=4, H=8, halfway = 6 = F)

15. Q (+2 each time)

Standard

1. K (+4 each time, wrapping from Z back round to A)

2. O (−4 each time, wrapping from A back round to Z)

3. TPERX (P→T, L→P, A→E, N→R, T→X)

4. A (L is the 12th letter; 12+15=27, which wraps round to 1 = A)

5. R (D is the 4th letter; 4−12=−8, which wraps round to R)

6. B (the 23rd letter is W; W+5 wraps round to B)

7. F (+3 each time, wrapping from Z back round to A)

8. PLUM (shift each letter of the coded word back by 5, wrapping round)

9. N (the gaps go +2, +3, +2, +3, so the next gap is +2: L + 2 = N)

10. O (G=7, W=23, halfway = 15 = O)

11. YIZEV (B ↔ Y, R ↔ I, A ↔ Z, V ↔ E, E ↔ V)

12. BRAVE (the reverse-code swap undoes itself)

13. P (+6 each time, wrapping from Z back round to A)

14. F (32 wraps round: 32−26=6 = F)

15. K (the code is its own reverse, so the letter that swaps to P is the same letter P swaps to: K)

Standard / Advanced

1. E (two sequences woven together: 1st/3rd/5th letters A, C, E go +2; 2nd/4th letters Y, W go −2)

2. L (the gap changes partway: +2, +2, then +3)

3. D (+2 each time, wrapping from Z back to B)

4. FQI (working out the hidden shift from CAT→ECV shows it is +2, not +1: D→F, O→Q, G→I)

5. O (the gaps follow 1, 1, 2, 3, so the next gap is 5)

6. AGNR (Y→A, E→G, L→N, P→R)

7. VCE (reverse to TAC, then T→V, A→C, C→E)

8. DCW (C+1=D, A+2=C, T+3=W)

9. V (the even-position letters run Z, X, ?, decreasing by 2: X − 2 = V)

10. SZMW (H→S, A→Z, N→M, D→W)

11. ZLQ (W→Z, I→L, N→Q)

12. R (gaps are +1, +2, +3, +4, so the next gap is +5: M + 5 = R)

13. PNO (C→P, A→N, B→O)

14. F (16 doubles to 32, which wraps to 32−26 = 6 = F)

15. NYOR (L→N, A→Y (wraps back past A to Y), M→O, P→R)